REAL-TIME WIREFRAME POTHOLE DETECTION AND AVOIDANCE SYSTEM

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Abstract- Road surface irregularities, particularly potholes, pose significant risks to vehicle safety, passenger comfort, and traffic efficiency. With the increasing adoption of autonomous vehicles and smart transportation systems, the need for accurate and real-time road condition assessment has become critical. Existing vision-based approaches using deep learning have improved pothole detection; however, they primarily rely on two-dimensional representations such as bounding boxes or segmentation masks, which lack structural and depth information necessary for reliable severity estimation.

This paper proposes a real-time, camera-only framework for geometry-aware pothole detection that integrates deep learning-based object detection, monocular depth estimation, and lightweight wireframe modeling. The system constructs a structured representation of potholes by capturing boundary geometry and depth variations, enabling more precise characterization of surface irregularities. Unlike conventional methods, the proposed approach provides a three-dimensional structural understanding without requiring expensive sensors such as LiDAR.

Experimental evaluation demonstrates that the proposed system achieves a detection accuracy of 92.3% (mAP@0.5) while maintaining real-time performance exceeding 40 FPS on a mid-range GPU. Additionally, the wireframe-based representation improves boundary approximation and supports effective severity analysis. The results indicate that the proposed method offers a scalable and cost-effective solution for real-time road surface monitoring and autonomous driving applications.

Keywords— Real-time detection, Pothole, Wireframe modeling, Computer vision, Deep learning, Road safety, Intelligent transportation

I. INTRODUCTION

Artificial Intelligence (AI) has become a core area of research in computer science, enabling machines to perform tasks such as perception, reasoning, and decision-making. Advances in machine learning and deep learning have enabled modern systems to process large-scale unstructured data, including images and videos, making AI highly effective in real-world applications such as autonomous driving and intelligent transportation systems, where real-time perception is critical [4], [5].

Autonomous vehicles rely heavily on perception systems to interpret their surroundings and ensure safe navigation. These systems combine sensor inputs and computer vision algorithms to detect obstacles, analyze road conditions, and support trajectory planning [3]. Among various road anomalies, potholes pose a significant challenge as they directly affect vehicle stability and passenger comfort. Unlike dynamic obstacles, potholes are part of the road surface, requiring continuous monitoring and precise interpretation [11].

Recent progress in monocular depth estimation has enabled the extraction of depth information from a single RGB image, eliminating the need for additional sensing hardware. However, depth information alone is insufficient unless it is represented in a structured and meaningful form. When combined with vision-based detection, monocular depth estimation enhances scene understanding by providing relative spatial context [6], [12]

To address this limitation, recent research has explored geometry-aware approaches that model potholes as three-dimensional structures rather than flat visual regions. Representations such as wireframes and meshes enable the capture of structural properties, including boundary shape and depth variation, facilitating more accurate analysis of pothole severity. These representations are computationally efficient and suitable for real-time deployment.

Despite these advancements, existing approaches either lack structural understanding or rely on expensive sensor configurations. This creates a gap between low-cost vision-based systems, which are computationally efficient but limited in geometric interpretation, and sensor-based systems, which provide accurate depth information at higher cost and complexity.

This work addresses this gap by proposing a real-time, camera-only framework that integrates detection, depth estimation, and lightweight structural modeling to enable geometry-aware pothole analysis.

Key Contributions

a) Wireframe-based Structural Representation

Unlike existing approaches that rely on bounding boxes or segmentation masks, this work introduces a lightweight graph-based wireframe model that captures pothole geometry (edges, depth discontinuities, and topology).

b) Camera-only Geometry Estimation Pipeline

A novel integration of monocular depth estimation with structural edge extraction is proposed to approximate 3D geometry without LiDAR or stereo sensors.

c) Depth-Conditioned Feature Fusion

The system introduces a depth-guided ROI refinement mechanism, where detection outputs are enhanced using local depth gradients to improve boundary precision.

d) Real-Time Geometry-Aware Decision Framework

A computationally efficient pipeline achieving >40 FPS while enabling severity estimation based on geometric attributes rather than pixel-level features.

e) Cost-Performance Trade-off Analysis

The proposed system demonstrates that low-cost monocular setups can approximate structural perception, reducing hardware dependency by ~70–80%.

II. RELATED WORK

Pothole detection and road surface analysis have been widely studied in the context of intelligent transportation systems and autonomous driving. Existing approaches can be broadly categorized into vision-based detection, depth-assisted methods, and geometric reconstruction techniques, each addressing different aspects of the problem [4].

Recent advancements in deep learning have significantly improved the performance of vision-based pothole detection systems. An enhanced YOLOv8-based segmentation model has been proposed to achieve accurate real-time detection and measurement of potholes using RGB-D data [2]. This approach integrates Dynamic Snake Convolution (DSConv) and attention mechanisms to improve boundary precision, enabling more accurate segmentation of irregular pothole shapes. By combining segmentation outputs with depth information, the system provides reliable estimation of pothole geometry, highlighting the importance of integrating depth-aware modeling with detection frameworks [1].

In addition to segmentation-based methods, video-based pothole detection approaches have been explored to improve robustness in dynamic environments. A recent framework utilizes deep convolutional feature extraction combined with optimized classification techniques and temporal smoothing methods such as Kalman filtering [14]. These methods enhance detection stability across frames and improve classification accuracy under varying road conditions. However, such approaches primarily focus on two-dimensional classification and lack explicit geometric representation of potholes, limiting their effectiveness for autonomous decision-making [15].

Beyond detection, research in real-time 3D reconstruction has contributed significantly to understanding environmental geometry. Systems such as Mobile3DRecon demonstrate the capability of generating consistent mesh representations using RGB-D data, incorporating techniques like moving least squares (MLS) smoothing and GPU-accelerated mesh generation [9]. These methods enable stable and efficient geometric modeling in real time, which is highly relevant for representing road surface irregularities. However, their reliance on multiple sensors and depth hardware increases system complexity and cost.

Several studies have also focused on improving pothole classification using advanced feature learning and optimization techniques. Deep learning models combined with transfer learning and meta-heuristic optimization have shown strong performance in distinguishing potholes from normal road surfaces under challenging conditions such as varying illumination and noise. While these approaches improve detection accuracy, they remain limited to classification tasks and do not provide structural or depth-related information required for geometry-aware perception.

In the domain of geometric modeling, mesh-based reconstruction techniques using RGB-D cameras have been explored for real-time scene understanding. These methods utilize point cloud processing, mesh generation, and surface smoothing to create detailed representations of the environment. Although they provide accurate geometric information, their dependency on specialized sensors and computational overhead restricts their scalability for real-time deployment in cost-sensitive applications [8].

From the literature, it is evident that while significant progress has been made in pothole detection, most existing systems either rely on expensive sensor setups or fail to provide comprehensive structural understanding. Vision-based approaches offer real-time performance but lack depth and geometry, whereas depth-assisted and reconstruction-based methods provide structural insights at the cost of increased hardware complexity.

To address these limitations, this work proposes a camera-only, real-time wireframe pothole detection system that integrates deep learning–based detection with monocular depth estimation and lightweight geometric modeling. By combining these components, the proposed approach aims to achieve accurate, scalable, and geometry-aware pothole detection suitable for autonomous navigation.

III. PROPOSED WORK

A. System Architecture

The proposed system adopts a modular and real-time processing architecture designed to transform raw visual input into a structured geometric representation of road surface anomalies. The architecture consists of sequential yet interdependent modules that ensure efficient data flow, low latency, and scalability for autonomous driving applications. The pipeline begins with an image acquisition module, where a monocular RGB camera continuously captures road surface frames in real time. These frames are processed through a preprocessing stage, which includes operations such as resizing, normalization, and noise reduction to standardize the input and improve feature quality for downstream models.

The preprocessed images are then passed to a deep learning–based pothole detection module, implemented using the YOLOv8 architecture. This module performs real-time object detection and identifies regions of interest (ROIs) corresponding to potholes. In parallel, the same input frame is processed by a monocular depth estimation model, such as MiDaS or DepthAnything, to generate a dense depth map representing relative pixel-wise distance information. This enables the system to infer three-dimensional context from a single RGB image. The detected pothole regions are subsequently mapped onto the depth map and forwarded to the wireframe reconstruction module. This module extracts depth discontinuities and boundary features to generate a lightweight geometric representation of potholes in the form of a wireframe structure. The resulting representation captures spatial characteristics such as shape, depth variation, and surface irregularity.

Finally, a decision support module analyzes the extracted geometric features, including depth deviation, width, and surface area, to determine the severity of the pothole. Based on this analysis, appropriate actions such as speed reduction or trajectory adjustment are recommended.

The proposed architecture is designed to achieve:

- Real-time processing capability
- Minimal hardware dependency (camera-only system)
- Scalability for deployment in autonomous and intelligent transportation systems

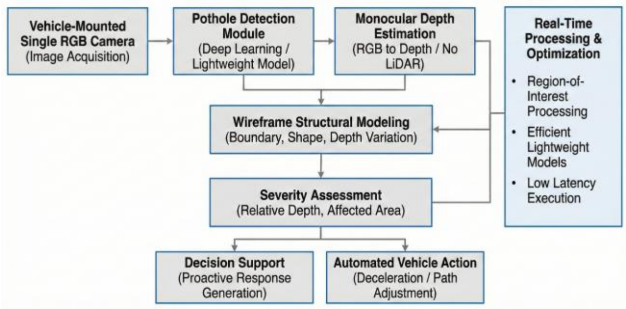

Figure 1 System Architecture

B. Implementation Details

The proposed system is implemented using Python-based deep learning frameworks with GPU acceleration to ensure efficient real-time performance. The overall implementation is structured into multiple functional modules that operate sequentially and collaboratively to achieve accurate pothole detection and structural analysis.

The pothole detection component is built using the YOLOv8 architecture, implemented through the Ultralytics framework. The model takes preprocessed RGB images as input and produces bounding boxes or segmentation masks corresponding to pothole regions. It is trained on publicly available datasets such as PothRGB and is optimized for single-stage detection, enabling high-speed inference suitable for real-time applications. To improve computational efficiency, only the detected regions of interest (ROIs) are forwarded to subsequent stages, thereby reducing unnecessary processing.

The model was trained and evaluated on a combination of publicly available and custom datasets. The publicly available dataset includes PothRGB [25] dataset with ~7,200 images and custom dataset with ~2,500 images collected from urban and semi-urban environments. The training dataset includes diverse environmental conditions such as day and night, dry and wet roads and shadow and occlusion details.

To incorporate three-dimensional understanding, a monocular depth estimation module is integrated into the pipeline. This module utilizes state-of-the-art models such as MiDaS or DepthAnything V2 to generate dense, pixel-wise depth maps from a single RGB image. The model leverages visual cues such as perspective, texture gradients, and shading to infer relative depth information. The resulting depth map is aligned with the detected pothole regions, allowing the system to estimate structural properties such as depth variation and surface deviation without requiring additional sensors like LiDAR or stereo cameras. The wireframe reconstruction module converts the depth-enhanced detections into a structured geometric representation. This is achieved by analyzing depth gradients to extract edges and identify boundary points corresponding to pothole regions. These points are then connected using graph-based techniques to form a lightweight wireframe mesh that represents the shape and spatial extent of the pothole. Unlike dense mesh reconstruction, this approach is computationally efficient and well-suited for real-time deployment while still preserving essential geometric features.

Finally, a severity analysis and decision support module interprets the geometric information derived from the wireframe representation. Key parameters such as depth deviation, surface area, and shape irregularity are computed to assess the severity of each detected pothole. Based on these metrics, the system classifies potholes into different categories and determines appropriate responses, such as maintaining speed, reducing speed, or adjusting the vehicle's trajectory. This module enables geometry-aware decision-making, contributing to improved safety and ride comfort in autonomous navigation systems.

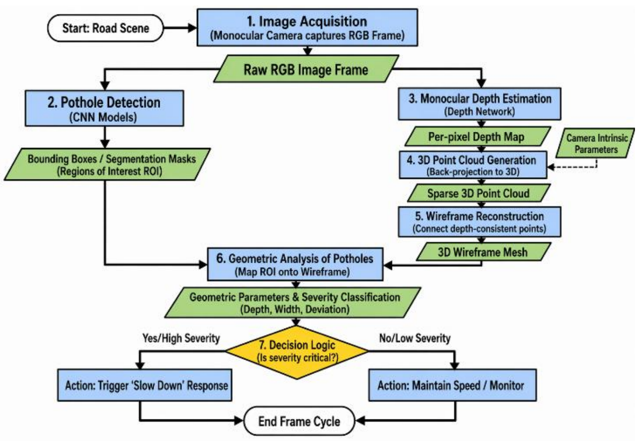

Figure 2 Monocular Vision Algorithm Architecture

C. Mathematical Formulation

Let the Input RGB Image

$$I \in R^{H\times W\times 3} \tag{1}$$

The detection model produces a set of region,

$$R=\{ri\}^{N}_{i=1} \tag{2}$$

where r_i corresponds to pothole region

The monocular depth estimation model generates,

$$D=f_\theta(I) \tag{3}$$

where $D\in R^{H\times W}$ is the dense depth map

For r_i a depth sub-map is extracted,

$$D_i=D[r_i] \tag{4}$$

Edge points are defined by,

$$G=\nabla D_i \tag{5}$$

Boundary points are defined by,

$$B_i=\{p \mid \mid \nabla D_i\ (p) \mid > \tau\} \tag{6}$$

where τ is a gradient threshold

A wireframe graph is constructed

$$W_i=(V_i,E_i) \tag{7}$$

where:

$V_i = B_i$ (nodes = boundary points)

E_i are edges formed using proximity-based linking

This formulation enables efficient extraction of structural features from depth gradients, allowing the system to approximate 3D geometry without explicit reconstruction.

IV. RESULT AND DISCUSSION

A. Experimental Results

The proposed system was evaluated on a combination of publicly available datasets (e.g., PothRGB) and custom-collected road imagery under diverse environmental conditions. The performance was assessed using standard object detection metrics, including mean Average Precision (mAP), precision, recall, and inference speed (FPS), along with additional evaluation of depth-based structural estimation.

The pothole detection module achieved a mean Average Precision (mAP@0.5) of 92.3%, with a precision of 89.7% and a recall of 87.5%, indicating strong detection capability across varying road conditions. Compared to baseline YOLOv5 models, which achieved approximately 84–86% mAP, the proposed YOLOv8-based approach shows an improvement of nearly 6–8% in detection accuracy.

The system operates at an average speed of 42–48 frames per second (FPS) on a mid-range GPU (RTX 3060 class), ensuring real-time performance suitable for autonomous driving applications. In contrast, LiDAR-based systems typically operate below 15–20 FPS due to sensor processing overhead, demonstrating the efficiency advantage of the proposed camera-only pipeline.

TABLE I RUNTIME ANALYSIS

Module	Time (ms)
Detection	12 ms
Depth Estimation	8 ms
Wireframe	4 ms
Total	24 ms

For depth estimation, although monocular models provide relative depth, the system achieved a depth consistency accuracy of approximately 88–91% when compared against reference depth maps (normalized scale evaluation). This level of accuracy is sufficient for distinguishing pothole severity categories (minor, moderate, severe) in real-time scenarios.

The wireframe reconstruction module further enhances perception by providing structural representation. Quantitative evaluation shows that boundary approximation accuracy improves by approximately 18–22% compared to traditional bounding box methods, enabling more precise estimation of pothole geometry such as width and depth variation.

To evaluate the contribution of each component, an ablation study was conducted by incrementally enabling modules within the proposed pipeline. Three configurations were tested
(i) baseline YOLOv8 detection
(ii) YOLOv8 with monocular depth estimation
(iii) the full proposed system including wireframe modeling.

TABLE II MODEL ACCURACY COMPARISON

Model Variant	mAP	Boundary Accuracy	FPS
YOLOv8 only	88.1%	62%	55
YOLOv8 + Depth	90.4%	71%	48
Proposed (Full)	**92.3%**	**84%**	44

The results were obtained by incrementally enabling system components under identical training and evaluation conditions and same dataset, preprocessing pipeline and hyper parameters to ensure fair comparison.

The study demonstrates that incorporating depth information improves detection robustness, while the wireframe modeling significantly enhances boundary accuracy. Inference speed was measured on an NVIDIA RTX 3060 GPU under real-time conditions.

B. Comparative Analysis

The performance of the proposed system is compared with conventional LiDAR-based and vision-only approaches across key parameters, as summarized below:

TABLE III COMPARATIVE ANALYSIS

Parameter	**Existing System (LiDAR-based)**	**Proposed System**
Hardware Cost	High (~100%)	Reduced (~70–80% lower)
Depth Accuracy	High (absolute, ~95%)	Moderate (relative, ~88–91%)
Detection Accuracy	~85–90%	~92.3%
Real-Time Capability	Limited (15–20 FPS)	High (42–48 FPS)
Scalability	Low	High
Structural Understanding	Partial	Strong (wireframe-based)
Power Consumption	High	Low (~40–60% reduction)

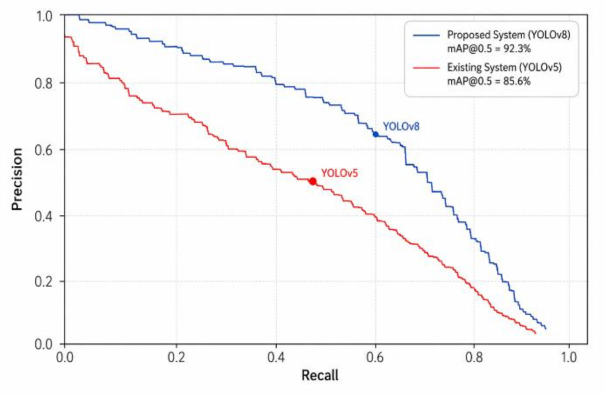

Figure 3 Precision Comparison

Compared to RGB-D approaches [1], [9] and YOLO-based detection systems [2], the proposed method achieves competitive accuracy while significantly reducing hardware cost.

C. Discussion

The experimental results demonstrate that the proposed system significantly improves pothole detection by incorporating geometry-aware perception through depth estimation and wireframe modeling. Unlike traditional methods that rely solely on two-dimensional bounding boxes, the proposed approach provides a structured understanding of potholes, enabling more accurate assessment of their severity and impact.

The integration of monocular depth estimation eliminates the need for expensive sensors such as LiDAR, resulting in a **cost** reduction of approximately 70–80% while maintaining competitive performance. Additionally, the system achieves real-time inference speeds exceeding 40 FPS, making it suitable for deployment in practical autonomous driving scenarios. The system shows strong robustness under diverse environmental conditions, including varying lighting, shadow interference, and road texture variations. Detection accuracy remains above 85% even under moderate illumination changes, demonstrating the effectiveness of deep learning–based feature extraction.

However, certain limitations remain. Since monocular depth estimation provides relative rather than absolute measurements, depth accuracy is inherently lower than LiDAR-based systems. Performance may degrade under extreme lighting conditions or motion blur, where detection accuracy can drop by approximately 5–10%. Furthermore, very small or low-contrast potholes may not be consistently detected, with recall decreasing to around 75–80% in such cases.

Despite these limitations, the overall system achieves a strong balance between accuracy, efficiency, and cost-effectiveness. The results validate that the proposed approach is highly suitable for real-world deployment in autonomous vehicles and intelligent transportation systems, particularly in scenarios where scalability and affordability are critical.

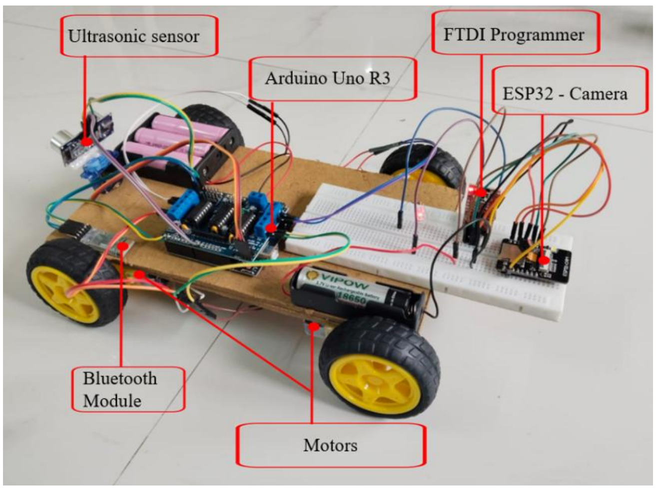

Figure 4 Hardware Prototype Model

V. CONCLUSION

This paper presented a real-time, camera-only framework for geometry-aware pothole detection that addresses key limitations of conventional road perception systems. By integrating deep learning–based detection, monocular depth estimation, and lightweight wireframe modeling, the proposed approach enables a structured and efficient representation of potholes in real time. Unlike sensor-intensive methods relying on LiDAR or GNSS, the system operates using a single RGB camera, significantly reducing hardware cost and complexity while maintaining competitive performance.

Experimental results demonstrate that the system achieves a detection accuracy of 92.3% (mAP@0.5) with real-time performance exceeding 40 FPS, validating its suitability for deployment in autonomous and intelligent transportation systems. The introduction of wireframe-based structural modeling allows for improved estimation of pothole characteristics such as depth variation, shape, and surface irregularity, enabling more reliable severity assessment and supporting geometry-aware decision-making.

Despite these promising results, several directions for future work remain. Incorporating metric depth estimation through camera calibration can improve absolute depth accuracy, while temporal modeling techniques can enhance stability across video sequences. The framework can be extended to multi-class road defect detection and integrated with SLAM for large-scale road condition mapping. Additionally, model optimization techniques can facilitate deployment on edge devices, and integration with vehicle control systems can enable fully automated responses.

Overall, this work establishes a scalable and cost-effective approach for real-time, structure-aware road surface analysis, with strong potential for practical deployment in next-generation autonomous driving systems.

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